

TRIZ Principles

Lecture 9

Outline

- What is TRIZ?
- Technical contradictions
- Physical contradictions
- Classroom Activity – Fusion 360:
 - Extruding, revolving, sweeping, creating circular patterns or holes in solid bodies
 - Working with multiple parts
 - Applying joints to components and animating them

TRIZ

- A Russian acronym:

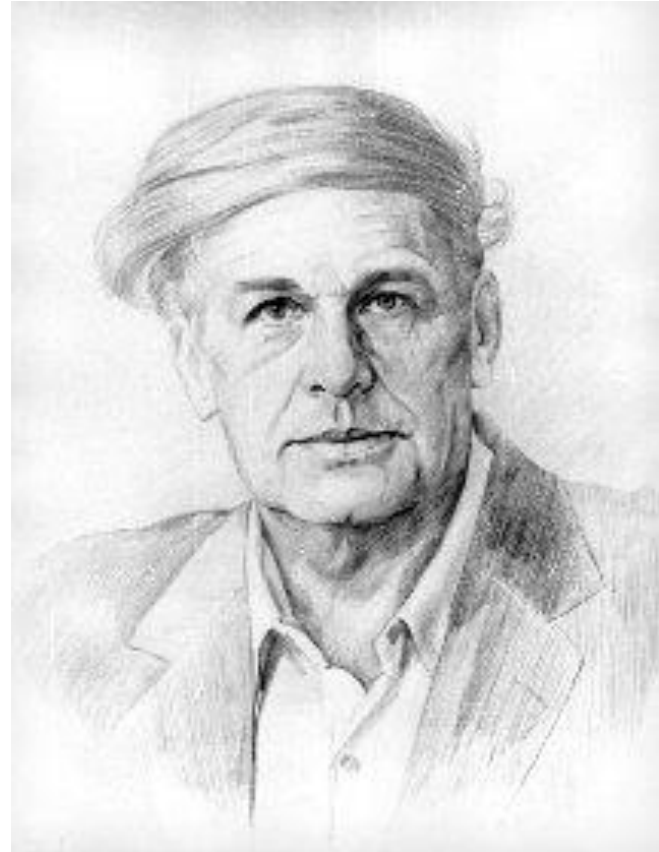
Theoria Resheneyva Isobretatelskehuh Zadach

ТЕОРИЯ РЕШЕНИЯ ИЗОБРЕТАТЕЛЬСКИХ ЗАДАЧ

(Theory of Inventive Problem Solving)

Who Developed TRIZ?

- Genrich Saulovich Altshuller (1926-1998) was born in Tashkent, Uzbekistan.
- 1946, working in Soviet Navy patent office
- He studied more than 400,000 worldwide patents to create the Contradictions Matrix with 40 inventive principles.
- 1948, wrote a letter to Comrade Stalin wishing to help the motherland create better inventions
- 1950, arrested for “sabotage” and sent to the Gulag
- 1956, wrote his first paper.

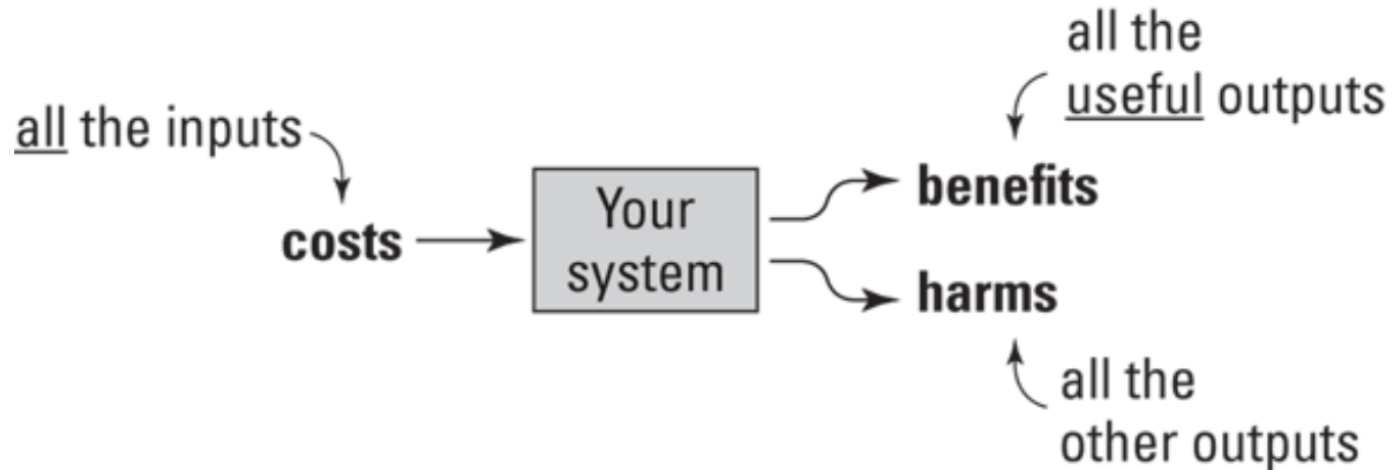


What is TRIZ?

- TRIZ is an evolving, open-ended system for enhancing human inventiveness through the following:
 - Systematic identification of problems and ideal solutions
 - Overcoming various blocks through heuristics and approaches that have worked in other disciplines
- TRIZ is based on two basic principles
 - Somebody, sometime, somewhere has already solved your problem or one similar to it.
 - Creativity means finding that solution and adapting it to the current problem.
 - Don't accept contradictions. Resolve them.

Ideality

$$\text{Ideality} = \frac{\uparrow \text{benefits}}{\downarrow \text{costs} + \text{harms} \downarrow}$$

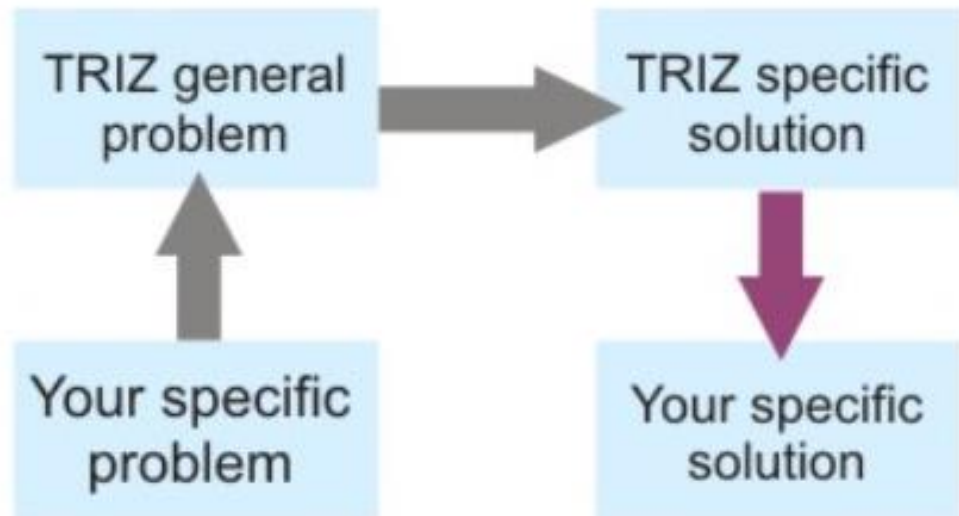


Ideal (To Be)

Difference (Gap) = Problem

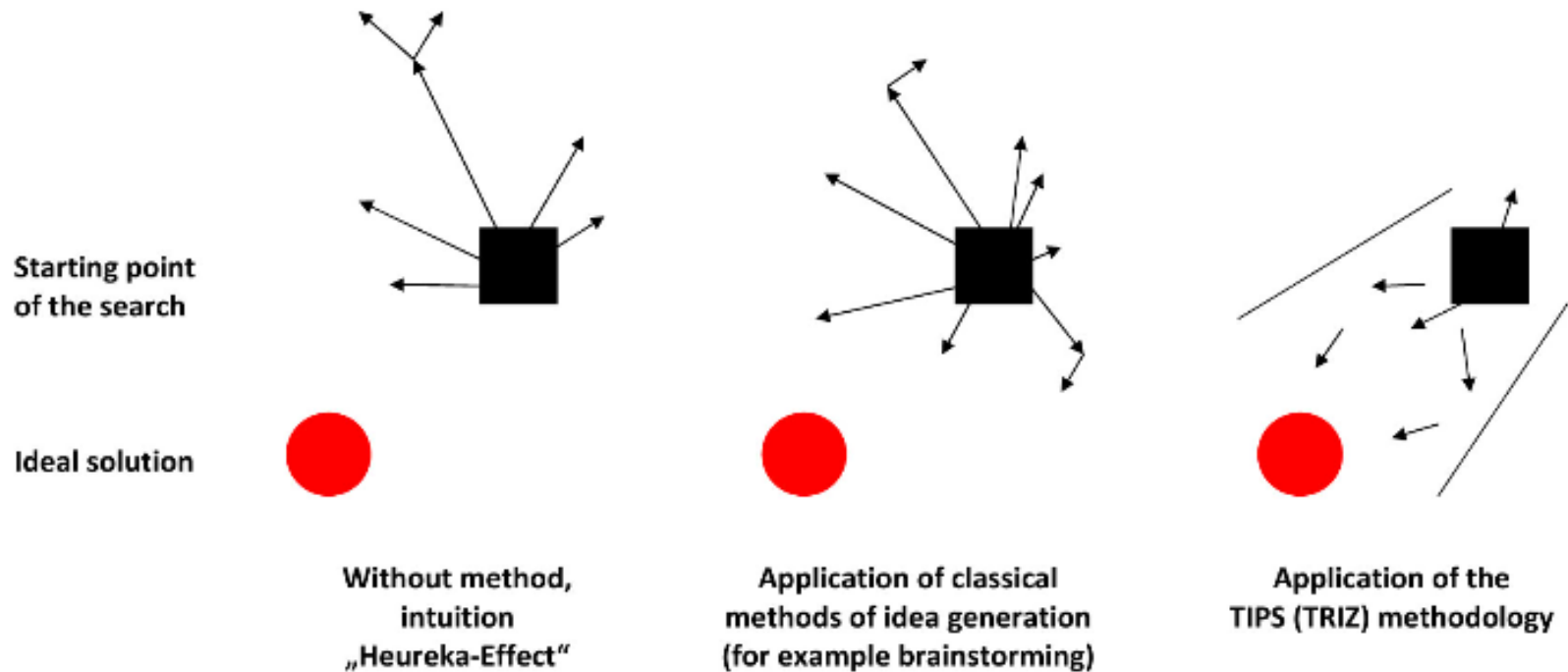
Present (As Is)

**TRIZ Problem-
Solving Method**

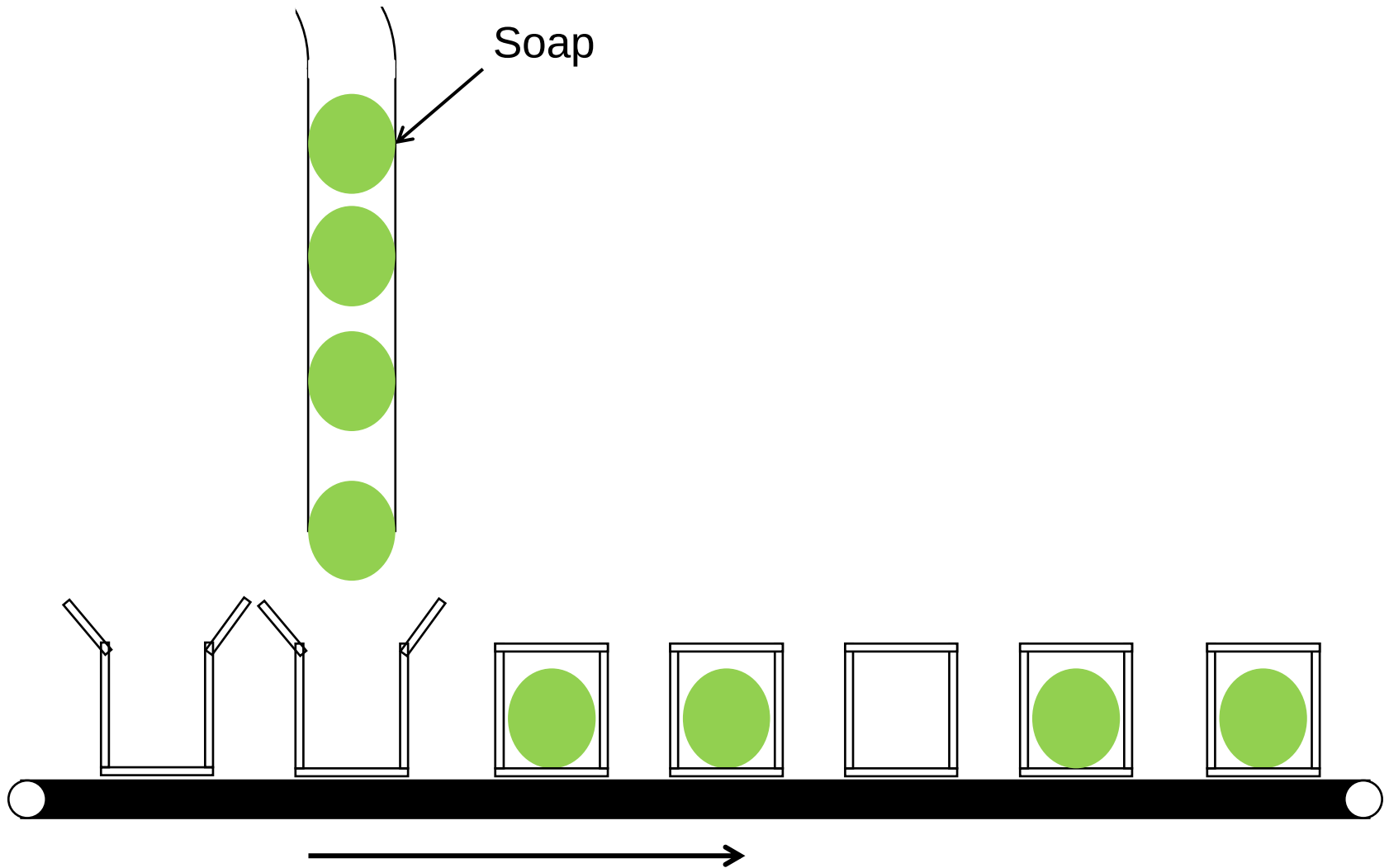


Finding Solution

- Finding solution with and without TRIZ



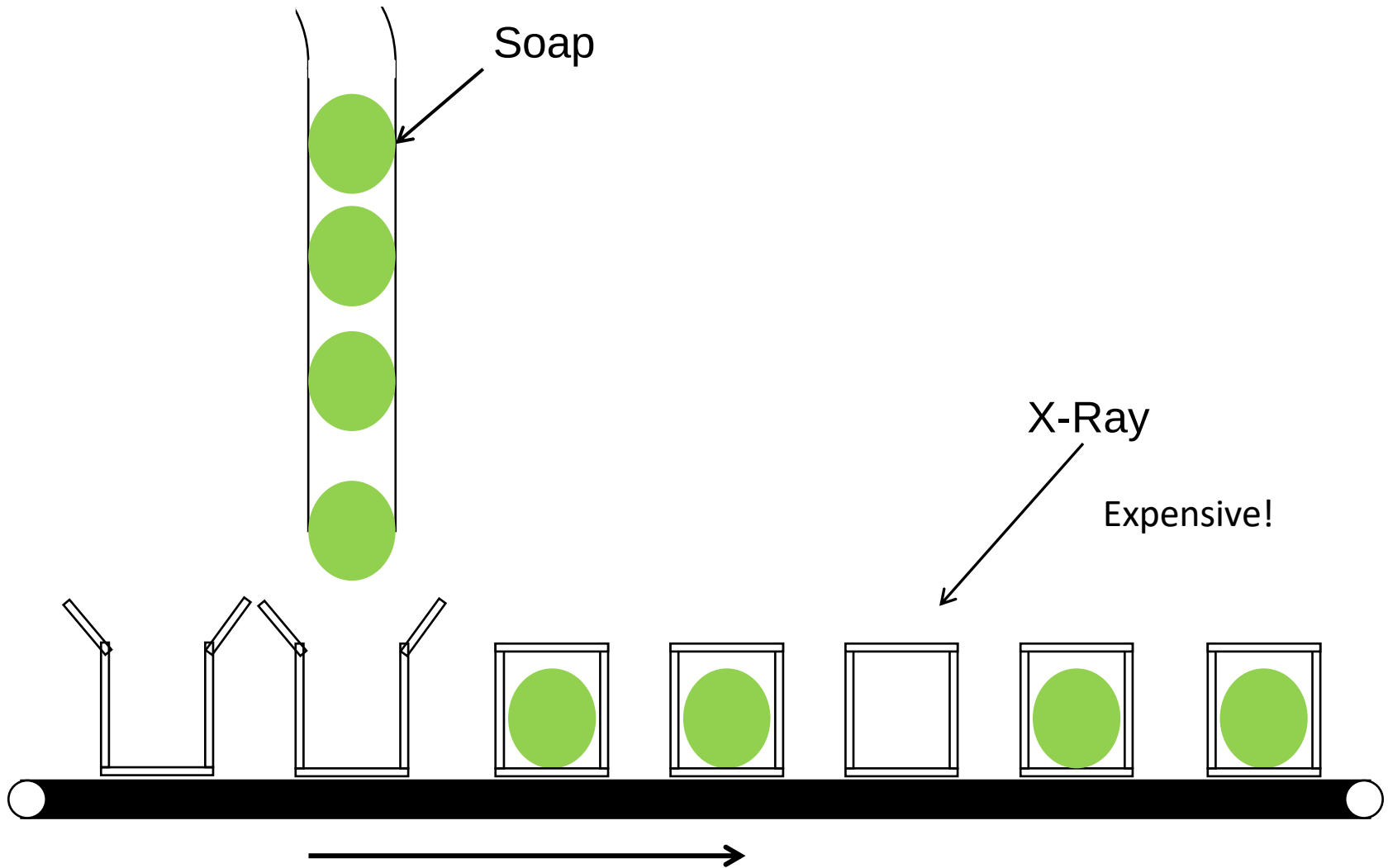
Soap Defect



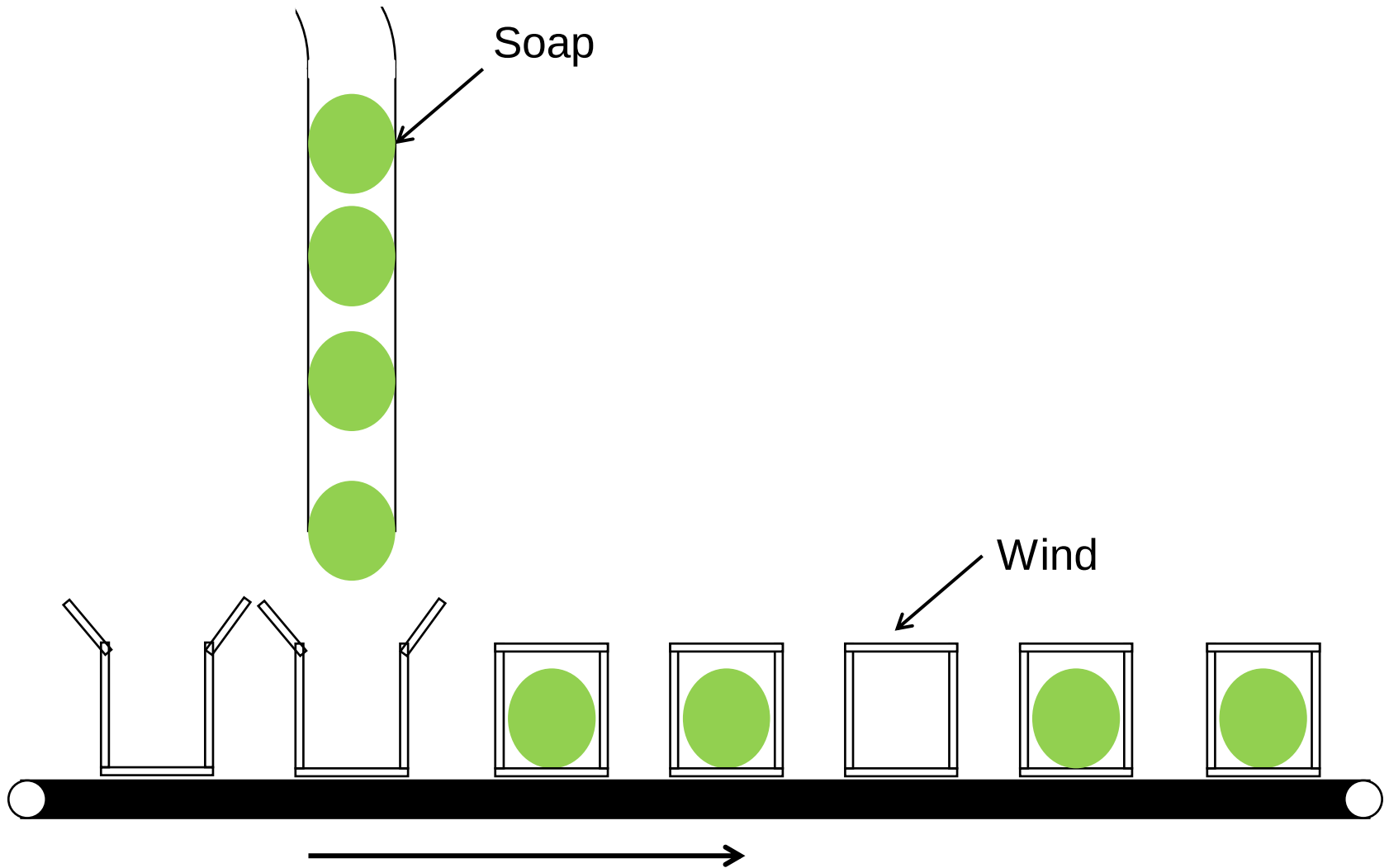
Empty Bath Soap Boxes

- What is the IDEAL FINAL RESULT?
 - We do not want to ship any empty soap boxes
- What are the RESOURCES we have?
 - Can we use the defect as a resource to solve the problem?

Soap Defect



Soap Defect

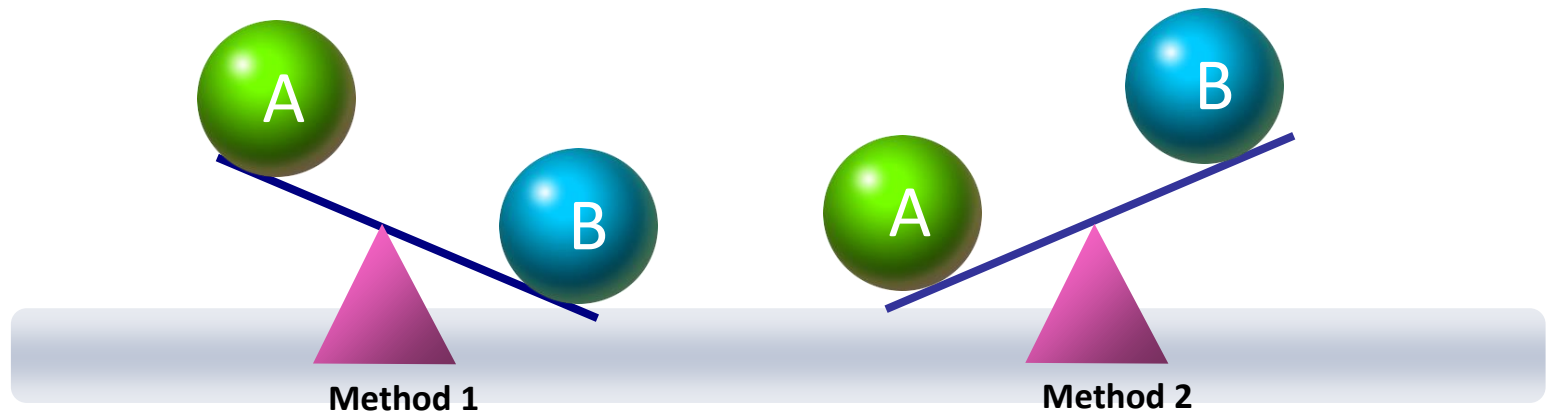


Contradiction

- **Technical Contradiction**

- TC: when we improve **parameter** A, then parameter B becomes worse.
- All general parameters can be represented by **39 technical parameters**.
- To solve the contradiction, **40 inventive principles** are recommended.
- 40 inventive principles came from patent investigations, and they helped solve the contradiction without any compromises.

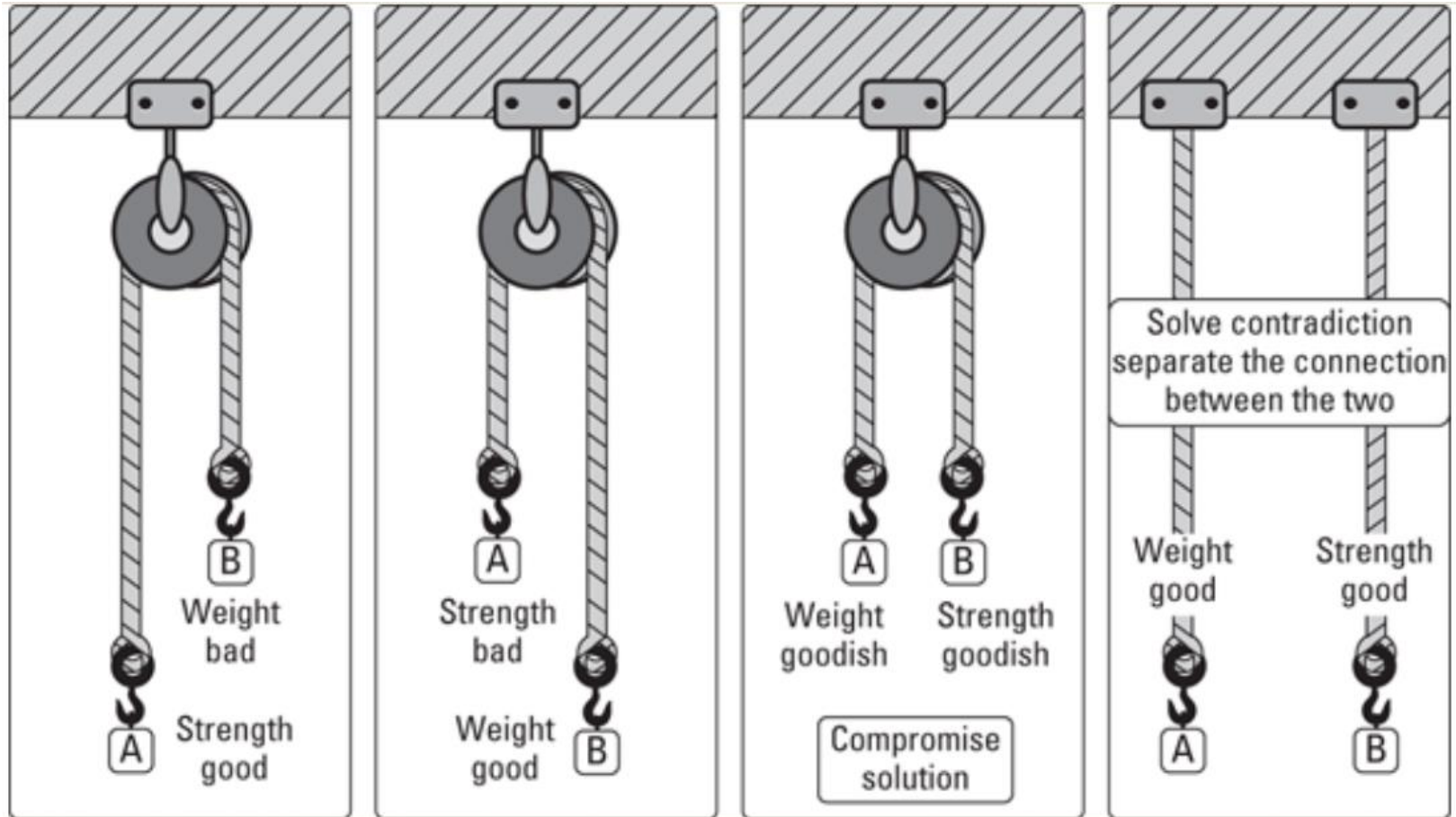
Technical Contradiction



TC 1: If we apply 'Method 1' then, <Property A> will grow better (good),
but <Property B> will grow worse (bad).

TC 1: If we apply 'Method 2' then, <Property B> will grow better (good),
but <Property A> will grow worse (bad).

Remove connection between A and B



How to Solve TC?

**Define
Contradiction**



**Parameter
Changing**



**Contradiction
Matrix**



**40 Inventive
Principles**

- **TC1:** If we perform an action, Parameter **A** grows better, Parameter **B** grows worse.
- **TC2:** If we perform an action, Parameter **B** grows better, Parameter **A** grows worse.

Changing A, B parameters into 39 technical parameters

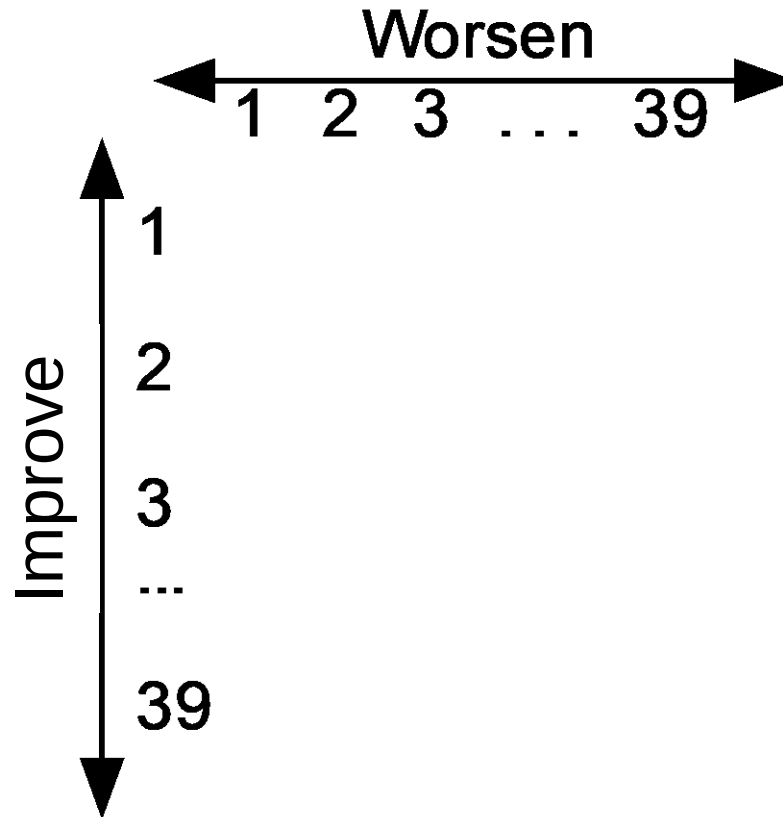
Using a contradiction matrix for selecting inventive principles

Idea generation from selected inventive principles

39 Technical Parameters

1: Weight of moving object	14: Strength	27: Reliability
2: Weight of stationary object	15: Durability of moving object	28: Measurement accuracy
3: Length of moving object	16: Durability of non moving object	29: Manufacturing precision
4: Length of stationary object	17: Temperature	30: Object-affected harmful
5: Area of moving object	18: Illumination intensity	31: Object-generated harmful
6: Area of stationary object	19: Use of energy by moving object	32: Ease of manufacture
7: Volume of moving object	20: Use of energy by stationary object	33: Ease of operation
8: Volume of stationary object	21: Power	34: Ease of repair
9: Speed of object	22: Loss of energy	35: Adaptability or versatility
10: Force (Intensity)	23: Loss of substance	36: Device complexity
11: Stress or pressure	24: Loss of Information	37: Difficulty of detecting
12: Shape	25: Loss of Time	38: Extent of automation
13: Stability of the object	26: Quantity of substance	39: Productivity

39 x 39 Contradiction Matrix



Contradiction Matrix

Worsening Feature →

→ **Improving Feature**

Worsening Parameter

Improving Parameter

8. Volume of Stationary Object

14. Strength

9, 14
17, 15

Suggested **inventive principles** that have solved similar contradictions before

40 Inventive Principles

- | | |
|----------------------------------|---|
| 1) Segmentation | 21) Skipping |
| 2) Separation | 22) Blessing in disguise |
| 3) Local quality | 23) Feedback |
| 4) Symmetry changes | 24) Intermediary |
| 5) Merging | 25) Self-service |
| 6) Multi-functionality | 26) Copying |
| 7) Nested doll | 27) Cheap disposables |
| 8) Weight compensation | 28) Mechanical interaction substitution |
| 9) Preliminary anti-action | 29) Pneumatics and hydraulics |
| 10) Preliminary action | 30) Flexible shells and thin films |
| 11) Beforehand compensation | 31) Porous materials |
| 12) Equip-potentiality | 32) Color changes |
| 13) The other way around | 33) Homogeneity |
| 14) Curvature increase | 34) Discarding and recovering |
| 15) Dynamic parts | 35) Parameter changes |
| 16) Partial or excessive actions | 36) Phase transitions |
| 17) Dimensionality change | 37) Thermal expansion |
| 18) Mechanical vibration | 38) Strong oxidants |
| 19) Periodic action | 39) Inert atmosphere |
| 20) Continuity of useful action | 40) Composite materials |

Example

- Make a cheese grater that can grate cheese quickly.
- Make it bigger?
 - Cheese goes all over the kitchen.
- These two parameters are fundamentally linked:
 - as the size of the grater decreases, control over where the cheese goes increases,
 - but the rate at which you can grate cheese drops.
- A conventional approach to this problem is to find a reasonable middle ground

TRIZ Approach

- Domain Specific Technical Contradictions:
 - A: Grating speed
 - B: Surface area

<i>What's getting better?</i>	<i>What's getting worse?</i>
Faster cheese grating	Lost cheese
TRIZ Technical Parameter: 39. Productivity	TRIZ Technical Parameter: 6. Area of stationary object

Parameter Resolution

- 7 – Nested Doll
- 10 – Prior Action
- 35 – Parameter Change
- 17 – Another Dimension

		Worsening feature Improving feature							
		Weight of moving object	Weight of stationary object	Length of moving object	Length of stationary object	Area of moving object	Area of stationary object	Volume of moving object	Volume of stationary object
		1	2	3	4	5	6	7	8
37	Difficulty of detecting or measuring	27 26 28 13	6 13 28 1	16 17 26 24	26	2 13 18 17	2 39 30 16	29 1 4 16	2 18 26 31
38	Extent of automation	28 26 18 35	28 26 35 10	14 13 17 28	23	17 14 13		35 13 16	
39	Productivity	35 26 24 37	28 27 15 3	18 4 28 38	30 7 14 26	10 26 34 31	10 35 17 7	2 6 34 10	35 37 10 2

Without the area getting worse (6)

More productivity (39)

Suggested principles that have solved similar contradictions before

Concepts



Student Practice 1

- Which inventive principles are used?
 - Locate a noisy compressor outside the building where compressed air is used.
 - Use fiber optics or a light pipe to separate the hot light source from the location where light is needed.
 - Use the sound of a barking dog, without the dog, as a burglar alarm.
 - In order to advertise to many people, flew helium balloons into the air.
 - The new chair contains a smaller chair designed for children inside the bigger chair for adults.
 - The product, in the process of development, will be distorted five degrees to the right side. Therefore, we distorted the product five degrees to the left side before starting the process.

Student Practice 2

- Problem definition: Bullet proof vests should be strong, but not heavy.
- Step 1: Identify the contradiction(s).
 - (for students, the below should not be shown)
 - Strength (improves) versus weight (worsens)
- Step 2: Look at the list of 39 parameters and identify those important to your contradiction.
 - (for students the below should not be shown)
 - Strength: #14, Weight: #2

Student Practice 2

- Step 3: Identify which are improving parameters and which are worsening parameters.
 - (for students, the below should not be shown)
 - #14 Strength (improves) versus #2 weight (worsens)
- Step 4: Refer to the TRIZ Contradiction Matrix to learn which of Altshuller's Principles may be useful for this problem.
 - The intersection of Column 2 and Row 14 gives the following principles 1, 26, 27, 40

Answers

1. Segmentation

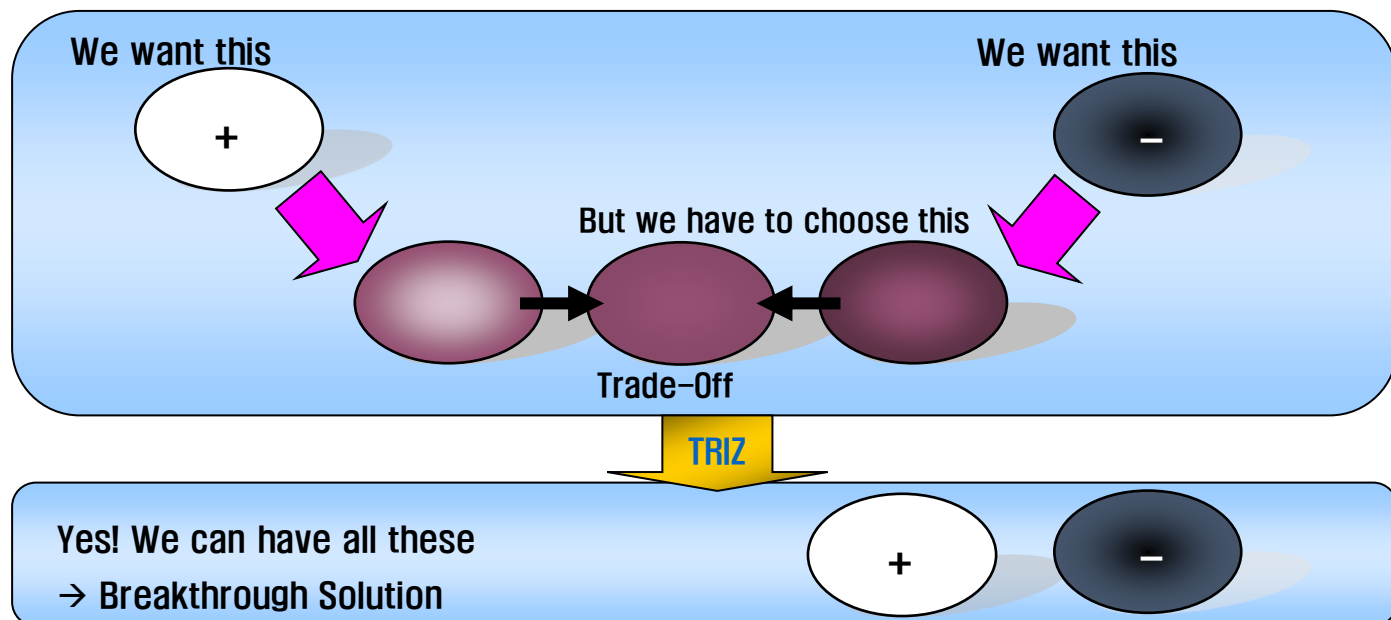
- Perhaps we could consider several different coverings for different parts of the body (pants, vest, etc.) rather than a one-piece suit.
- We could use different materials to cover the critical areas, such as chest and head, each taking advantage of the specific properties customized for the differing applications.

26. Copying

- Explanation of this Principle from the TRIZ website: Instead of an unavailable, expensive, fragile object, use simpler and inexpensive copies.
- Replace an object, or process with optical copies.

Physical Contradiction

- System parameter A must be positive (+) for one aspect of the system operation but it must be negative (-) for another aspect of the system operation.
 - Apply **separation principles**



Physical Contradictions

- Physical Contradictions are situations where one object has contradictory, opposite requirements.
- Everyday examples abound:
 - When pouring hot filling into chocolate candy shells, the filling should be hot to pour fast, but it should be cold to prevent melting the chocolate.
 - Aircrafts should be streamlined to fly fast, but they should have protrusions (landing gear) to maneuver on the ground.
 - Surveillance aircrafts should fly fast (to get to the destination) but should fly slowly to collect data directly over the target for long time periods.
 - Software should be easy to use, but should have many complex features and options.

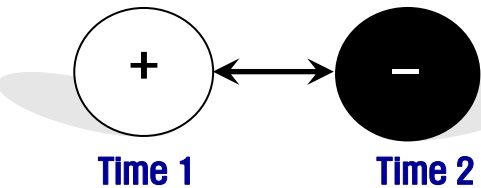
Physical Contradictions

- Automotive industry examples come from both design, production, and implementation:
 - Highways should be wide for easy traffic flow but narrow for low impact on communities.
 - Braking should be instantaneous to avoid road hazards but braking should be gradual for control.
 - Refueling should be sealed but also open.
 - The frame should be heavy (for structural safety) but also light (for cost and ease of assembly.)

Separation Principles

- **Separation In Time**
 - When it is time 1, parameter value is $+C$,
When it is time 2, parameter value is $-C$.
- **Separation In Space**
 - Where it is space 1, parameter value is $+C$,
Where it is space 2, parameter value is $-C$.
- **Separation In Condition**
 - When it is in condition 1, parameter value is $+C$,
When it is in condition 2, parameter value is $-C$.
- **Separation between the whole and its parts**
 - Partially, parameter value is $+C$, as a whole, parameter value is $-C$

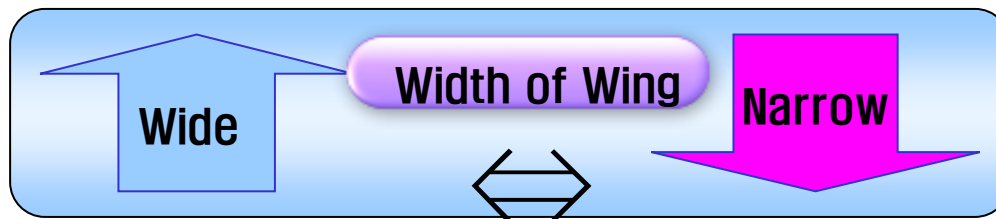
Separation Principle 1

No.	Separation Principle	Explanation	Visual Concept
1	Separation In Time	☞ When it is <Time 1>, <Property C in System> must be <+ Value>, when it is <Time 2>, < Same Property C in System > must be <- Value>. ☞ Required property is separated by time. ☞ <+> Property at time 1, <-> Property at time 2	

Example: Wing of an Airplane

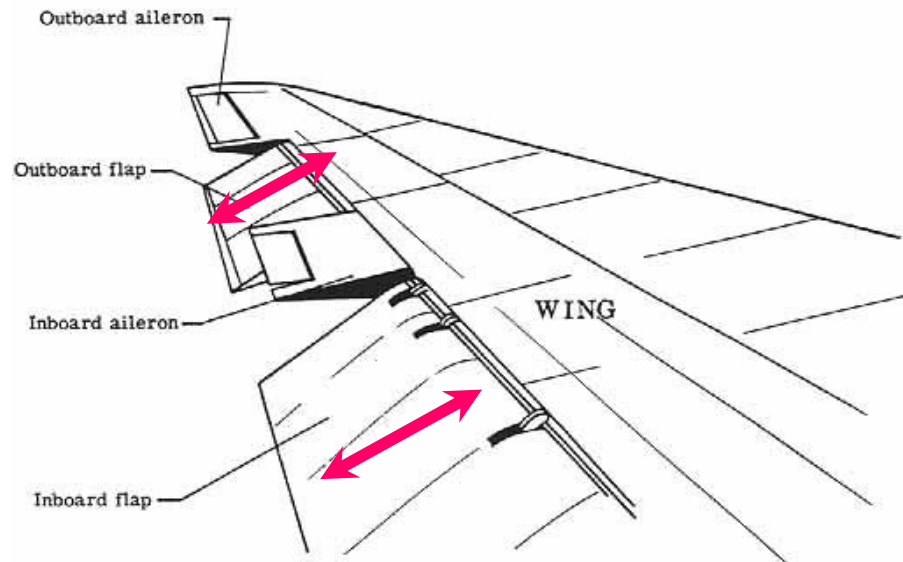
Physical Contradiction:

To take off, **the width of airplane must be wide**, and to speed up, **the width of airplane must be narrow**.



Separation Principle 1

Solution: When an airplane takes off (time 1), the width of the airplane becomes wide $<+>$; after taking off (time 2), the width of the airplane become narrow $<->$.



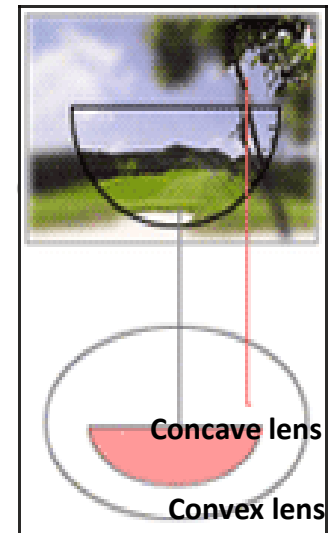
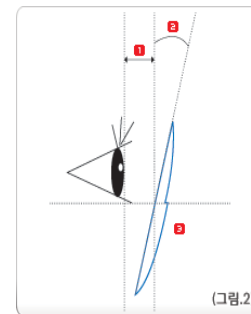
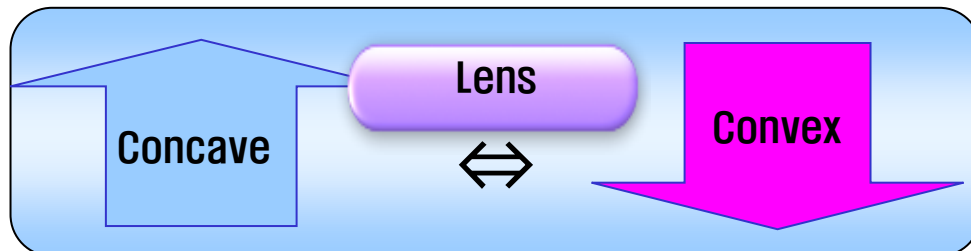
Separation Principle 2

No.	Separation Principle	Explanation	Visual Concept
2	Separation In Space	☞ When it is <Space 1>, <Property C in System> must be <+ Value>, when it is <Space 2>, < Same Property C in System > must be <- Value>. ☞ Exclusive required parameter is separated by space. ☞ <+> Property at one part, <-> Property at the other part	

Example: glasses for an old man

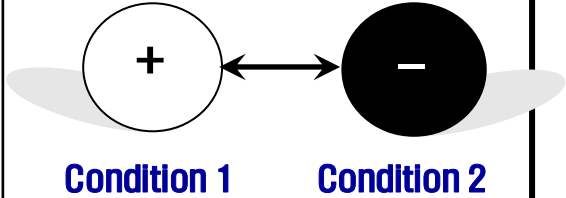
Physical Contradiction:

We want a concave lens for near-sighted eyes and a convex lens for long-sighted eyes.



“Multifocal point lens”

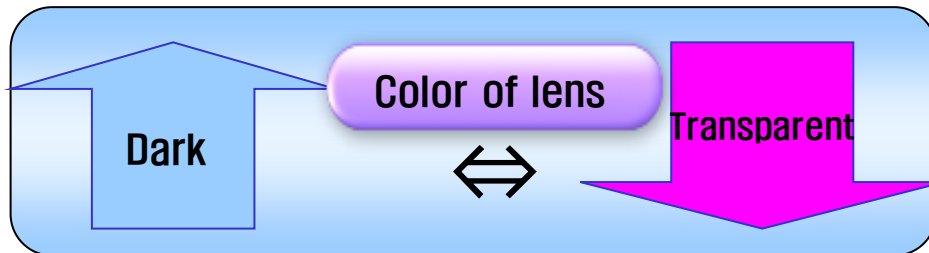
Separation Principle 3

No.	Separation Principle	Explanation	Visual Concept
3	Separation In Condition	- When it is <Condition 1>, <Property C in System> must be <+ Value>, when it is <Condition 2>, < Same Property C in System > must be <- Value>. - Exclusive required parameter is separated by condition. <+> Property at condition 1, <-> Property at condition 2	

Example: color changing sunglass

Physical Contradiction:

We want dark sunglasses outdoors but transparent lenses indoors.



"Ultraviolet reacting dye is coated on the lens"



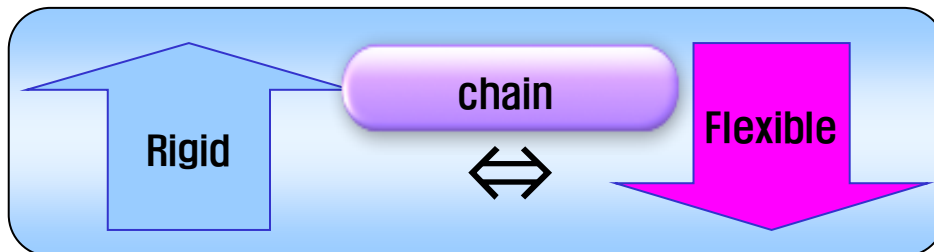
Separation Principle 4

No.	Separation Principle	Explanation	Visual Concept
4	Separation between the whole and its parts	☞ Partially <Property C in System> must be <+ Value>, totally < Same Property C in System > must be <- Value>. ☞ One parameter has one value in a general system and has the other value in partial system. ☞ Partially <+> Property, Totally <-> Property	

Example: chain in bicycle

Physical Problem:

Bicycle chains must be rigid and flexible for non-breakable transferring power.



Partially → Rigid
Wholly → Flexible



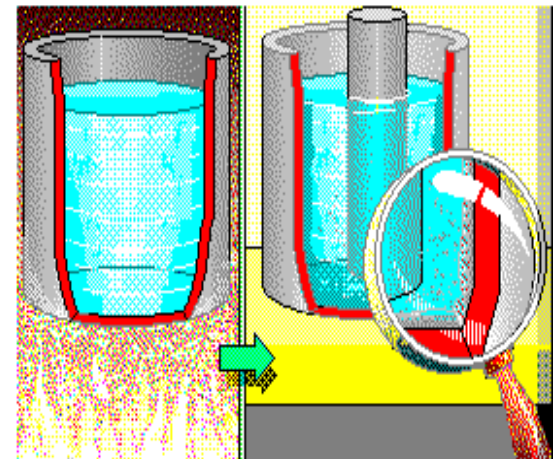
Yes y yes yesyes
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 Yes yes yes yes yes totally : No
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Problem 1

- **Coating metal workpieces**

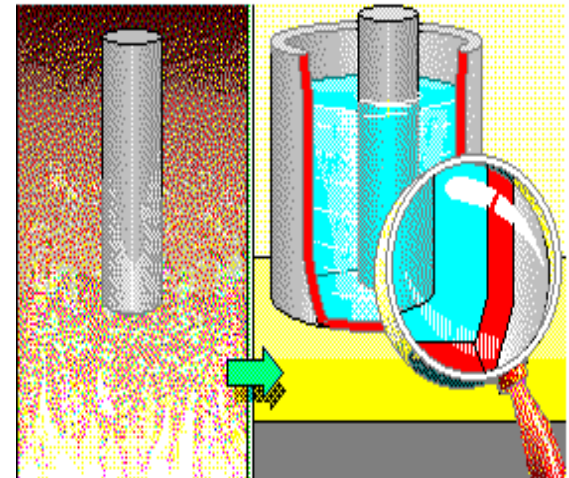
- Metal surfaces are chemically coated as follows: the metal workpiece is placed in a bath filled with a metal salt solution (e.g. nickel, cobalt, etc.). During the ensuing reduction reaction, metal from the solution precipitates onto the surface of the workpiece.

The higher the temperature, the faster the process takes place; however, at high temperatures the solution decomposes, and up to 75 percent of the chemicals are lost by settling on the bottom and sides of the bath. Adding stabilizers is not effective, and conducting the process at a low temperature sharply decreases production.



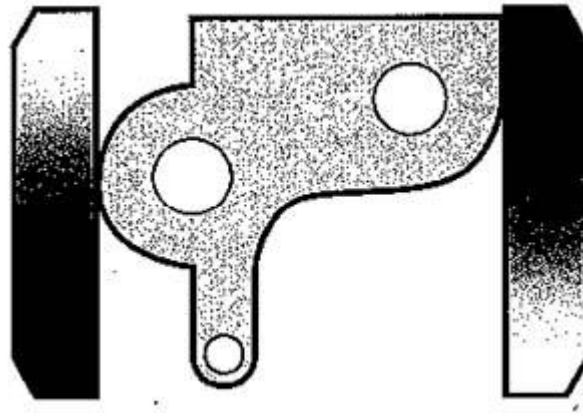
Answer to Problem 1

- The physical contradiction is as follows: The temperature should be high to increase productivity, and it should be low to avoid waste. To resolve physical contradictions, the following **separation principles** are recommended:
- To apply the principle of *separation in space*, for example, we should ask ourselves the following question: Do we need this parameter -- temperature, in this case -- to be high (and low) everywhere, or is it necessary in certain places only? If the temperature need not be both high and low everywhere, we can try to separate these opposite requirements in space.
- In this case, we need the temperature to be high *only near the parts* rather than everywhere in the bath. How can this be achieved?
- The answer is as follows:
- *The workpiece is heated to a high temperature before it is immersed in the solution, and the process itself is conducted at a low temperature. The solution is therefore hot near the workpiece but cold everywhere else. (One way to accomplish this is to apply an electric current to the workpiece during the coating process.)*



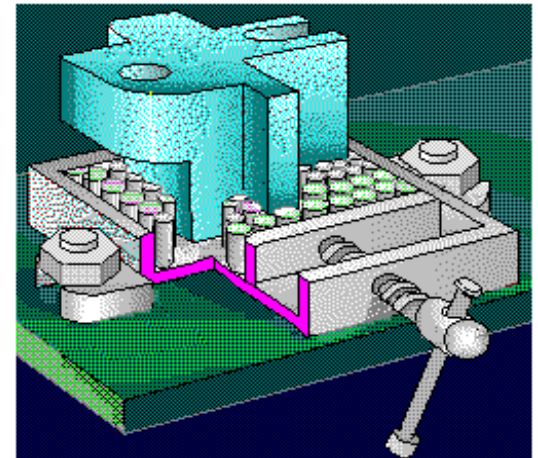
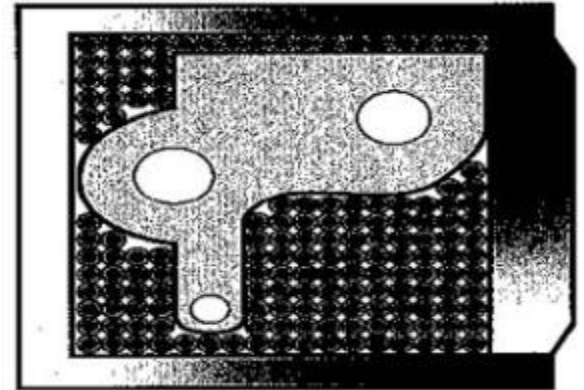
Problem 2

- **Gripping workpieces of complex shape**
 - To grip workpieces of complex shape, vice jaws must have a corresponding shape. It is expensive to produce a unique tool for every workpiece, however.



Answer to Problem 2

- The principle, **separation between the whole and its parts**, recommends that we assign one of the contradictory requirements to the whole system and the other to its parts:
- Use a vise with ordinary jaws, but add multiple hard bushings around the workpiece that move horizontally to conform to the workpiece's shape.
- US patent: 5145157, 5746423



Fusion 360